

Reviewer Pack — Minimal Rank of Quasi-crystalline Sheaves and DPLL Proof Tre...

Entry af9a1f2957c9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quasi-crystalline Sheaves and DPLL Proof Tree Height

Entry ID: af9a1f2957c9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 01:26:02 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Quasi-crystalline Sheaves) **Field B** (complexity object): Complexity Theory (DPLL Search Trees)

Statement:

For every Boolean satisfiability instance with n variables, the minimal rank of a quasi-crystalline sheaf associated with its Tseitin formula is linearly correlated with its DPLL proof tree height such that $\text{MinimalRank(QS)} = \Theta(\text{DPLLHeight})$.

Rationale (proposer's reasoning):

Quasi-crystalline sheaves provide a rich algebraic-geometric framework to study the structure of certain types of mathematical objects, and their ranks might encode essential complexity-theoretic information about DPLL proof trees, revealing deep connections between geometry and computational complexity.

Taxonomy category: Geometric Complexity Theory (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 2926eb090f277bef

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For the conjecture that $\text{MinimalRank(QS)} = \Theta(\text{DPLLHeight})$, a result will be considered supported if the Spearman's rank correlation coefficient (CRC) between the minimal ranks of quasi-crystalline sheaves and DPLL proof tree heights across 30 random seeds is ≥ 0.7 , and no seed results in a $\text{CRC} < 0.5$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def tseitin_encoding(formula):
        n = len(formula)
        new_vars = {}
        formulas_tseitin = []

        for i in range(n):
            if formula[i] not in new_vars:
                new_vars[formula[i]] = len(new_vars) + 1
            if -formula[i] not in new_vars:
                new_vars[-formula[i]] = len(new_vars) + 1

        def encode_clause(clause):
            literals = []
            for lit in clause:
                if lit > 0 and lit not in new_vars:
                    new_vars[lit] = len(new_vars) + 1
                elif lit < 0 and -lit not in new_vars:
                    new_vars[-lit] = len(new_vars) + 1
                literals.append(lit)
            return literals

        for clause in formula:
            literals = encode_clause(clause)
            if len(literals) == 2:
                formulas_tseitin.append([-new_vars[literals[0]], -new_vars[literals[1]], new_vars[literals[2]]])
                formulas_tseitin.append([new_vars[literals[0]], new_vars[literals[1]], -new_vars[literals[2]]])
            elif len(literals) == 3:
                formulas_tseitin.append([-new_vars[literals[0]], literals[1], literals[2]])
                formulas_tseitin.append([new_vars[literals[0]], -literals[1], literals[2]])
                formulas_tseitin.append([new_vars[literals[0]], literals[1], -literals[2]])
```

```

        formulas_tseitin.append([-new_vars[literals[0]], -literals[1], -literals[2]])

    return formulas_tseitin

def dpll(formula):
    n = len(formula)
    assignment = [None] * (n + 1)

    def solve(i):
        if i == n:
            return True
        for val in [True, False]:
            assignment[i + 1] = val
            if all(lit <= 0 or (assignment[abs(lit)] is not None and assignment[abs(lit)] == (lit > 0))):
                if solve(i + 1):
                    return True
            assignment[i + 1] = None
        return False

    return solve(0)

def minimal_rank(formula):
    n = len(formula)
    rank = 0
    for i in range(n):
        if all(lit <= 0 or (assignment[abs(lit)] is not None and assignment[abs(lit)] == (lit > 0))):
            rank += 1
    return rank

def spearman_correlation(ranks1, ranks2):
    n = len(ranks1)
    if n != len(ranks2):
        raise ValueError("Ranks lists must have the same length")

    sorted_ranks1 = sorted(range(n), key=lambda i: ranks1[i])
    sorted_ranks2 = sorted(range(n), key=lambda i: ranks2[i])

    rho_numerator = sum((sorted_ranks1[i] - sorted_ranks2[i]) ** 2 for i in range(n))
    rho_denominator = n * (n**2 - 1)

    return 1 - (6 * rho_numerator) / rho_denominator

n = random.randint(5, 40)
formula = [random.choice([i, -i]) for _ in range(n)]

formulas_tseitin = tseitin_encoding(formula)
dpll_height = len(dpll(formulas_tseitin))
minimal_rank_value = minimal_rank(formulas_tseitin)

return {
    "metric_name": "Spearman's rank correlation coefficient",
    "metric_value": spearman_correlation([i for i in range(n)], [minimal_rank_value] * n),
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

```

```

    }

if __name__ == "__main__":
    seeds = list(map(int, sys.argv[1:])) if len(sys.argv) > 1 else [2**i - 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(r["metric_value"] < 0.5 for r in results):
        first_failing_seed = next(i for i, r in enumerate(results) if r["metric_value"] < 0.5)
        print(f"RESULT: FALSIFIED counterexample='Spearman's rank correlation coefficient < 0.5'")
    else:
        print(f"RESULT: INCONCLUSIVE reason=insufficient_data n_tested={len(results)}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2f444def.py", line 114, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2f444def.py", line 94, in run_trial
    formula = [random.choice([i, -i]) for _ in range(n)]
                  ^
NameError: name 'i' is not defined. Did you mean: 'id'?

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating the conjecture's support conditions. | next: Debug the test code to ensure it runs successfully and produces the required data for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14032
2	preregistration	ollama_remote	glm4:latest	0	0	9767
3	novelty	ollama_remote	glm4:latest	0	0	10900
4	novelty	ollama_remote	glm4:latest	0	0	8791
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19960
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14657
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12143
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14457
9	judge	ollama_remote	glm4:latest	0	0	14733

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 119440 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/af9a1f2957c9.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/af9a1f2957c9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/af9a1f2957c9.tar.gz` (if generated)